

Meridian Parts

European service-hub expansion

Board decision report

Decision basis: 2025 shipped sales and returns known through 31 January 2026; archived ECB and World Bank public observations; synthetic hub-option policy.

Recommendation: conditionally fund NLD+ESP — Netherlands + Spain — with €270,000 year-zero capex and six FTE, subject to the 90-day gates in this report.

2025 net sales	2025 contribution	Recommended base increment	Defined-stress increment
€4,383,792	€2,030,010 (46.3%)	€269,815	€210,514

Prepared from the frozen source room; client data and scenario policy are synthetic. No live customer account or customer interviews were used.

Executive decision

Fund **NLD+ESP** conditionally, with a staged release of capex. It is the highest-stress feasible pair among alternatives that remain positive in low, base, high and stress cases. The board should approve the option envelope and the first 90 days, then release the second-hub commitment only if the measurable operating gates are met.

The recommendation gives up **€16,352** of modeled base annual increment versus the strongest base-only alternative (CZE+ESP), but improves the defined-stress result by **€90,267**. This is a risk preference under the client policy, not a proven forecast.

Scenario	Recommended NLD+ESP	CZE+ESP base leader	Payback NLD+ESP
Low	€134,532	€142,119	2.01x
Base	€269,815	€286,167	1.00x
High	€405,099	€430,215	0.67x
Stress	€210,514	€120,247	1.28x

Board asks

- Approve a €270,000 / six-FTE NLD+ESP envelope, with €180,000 capex and one FTE headroom remaining under policy.
- Authorize a 90-day validation sprint with a hard second-hub gate; do not treat the modeled payback as a guarantee.
- Require Finance, Operations and Data to sign off on actual lane-level SLA, return and contribution evidence before full release.

What would change the decision? A credible local-lane demand/uplift estimate below the low case, a site or labor quote that pushes capex above the envelope, or failure to meet the gate KPI plan would support deferring. If the board optimizes only for base-case arithmetic and accepts the modeled CZK exposure, CZE+ESP is the strongest alternative.

1. Scope, evidence and accounting

The model covers 2025 shipped orders, one SKU per order row, and returns received through the inclusive 31 January 2026 cutoff. It uses the highest numeric revision across both order extracts and corrections; identical repeats are one record. Returns are selected by return_id revision, additive across legitimate return IDs, and linked only to eligible orders. No orphan is matched by inference.

Evidence status

Client exports, corrections, costs, hub options and policy are synthetic source-room inputs. ECB reference-rate history and World Bank population/GDP are official public snapshots preserved in the source room, with original URLs and retrieval vintage. This report does not rely on live research, interviews or private customer data.

Check	Finding	Handling
Order rows	1,328 raw; 13 exact repeats; 18 revision replacements	Highest revision retained; no same-revision conflicts
Eligibility	1,254 eligible 2025 orders; exclusions {'shipped_after_2025': 1, 'status_not_shipped': 24, 'is_test': 18}	Cancelled/test/future records excluded
Returns	264 raw; 20 exact repeats; 241 included	One orphan and one after-cutoff row quarantined
Zero-price shipments	12 valid orders	Kept in orders/units; costs retained; no revenue invented
Reconciliation	72 country-month rows → six country totals	No monthly-to-country reconciliation issues

Revenue, cash and contribution

The analysis treats shipment-date gross sales as the booked-sales base and subtracts explicit refunds known by the cutoff to form net sales (€4,383,792). It does not reconstruct cash: payment settlement, receivables, taxes, inventory cash timing and refund settlement dates are not in the source room. Contribution (€2,030,010) is a management measure after net

COGS and non-refundable fulfillment, not cash and not accounting profit.

2. 2025 base diagnosis

Across the six markets, the retained base is 1,254 shipped orders and 77,436 units, producing €4,383,792 net sales and €2,030,010 contribution (46.3% margin). Spain has the highest contribution and net sales; Czechia has the next-highest contribution margin. Returns are physical units, not return-order counts, and the cutoff makes the comparison comparable across countries.

Market	Orders	Units	Net sales	Contribution	Margin	Returned units	Return rate
DEU	209	12,153	€625,792	€262,366	41.9%	73	0.6%
FRA	209	11,880	€632,231	€276,271	43.7%	144	1.2%
NLD	209	12,576	€697,733	€315,644	45.2%	144	1.1%
POL	209	13,163	€761,890	€356,494	46.8%	180	1.4%
CZE	209	13,733	€815,307	€397,335	48.7%	288	2.1%
ESP	209	13,931	€850,839	€421,900	49.6%	252	1.8%
Total	1,254	77,436	€4,383,792	€2,030,010	46.3%	1,081	1.4%

Financial bridge (EUR)

Market	Gross sales	Refunds	Net sales	Net COGS	Fulfillment	Contribution
DEU	€631,245	€5,453	€625,792	€349,980	€13,446	€262,366
FRA	€640,930	€8,699	€632,231	€341,453	€14,507	€276,271
NLD	€709,109	€11,376	€697,733	€365,496	€16,593	€315,644
POL	€776,508	€14,618	€761,890	€386,698	€18,698	€356,494
CZE	€834,572	€19,265	€815,307	€397,062	€20,910	€397,335
ESP	€872,259	€21,420	€850,839	€406,230	€22,708	€421,900
Total	€4,464,622	€80,831	€4,383,792	€2,246,919	€106,862	€2,030,010

Read-through

The country ranking is a diagnostic of this synthetic 2025 flow, not a causal estimate of hub demand. Spain and Czechia have the largest unit/contribution base, while Netherlands is smaller but has a favorable contribution profile and no PLN/CZK stress penalty. A hub case still depends on actual lane-level response, service-level improvement and the fixed-cost/fulfillment savings assumptions.

3. Market context and historical FX

The archived World Bank series show population change and current-US\$ GDP-per-capita context. Netherlands has the highest GDP per capita in 2024; Spain has the largest population growth among the six in this archive, while Poland is the only market with a population decline from 2022 to 2024. These are context signals only, not direct proof of product demand. Source vintage: 2026-07-13 in the archived response [S5–S6].

Market	Population 2022	Population 2024	Change	GDP pc 2024	GDP pc change
DEU	83,177,813	83,516,593	0.4%	\$56,103.73	11.1%
FRA	68,184,457	68,551,653	0.5%	\$46,103.08	12.5%
NLD	17,700,982	17,993,485	1.7%	\$67,465.32	14.1%
POL	36,821,749	36,559,233	-0.7%	\$25,103.57	32.9%
CZE	10,672,118	10,905,028	2.2%	\$31,827.76	12.5%
ESP	47,786,102	48,848,840	2.2%	\$35,326.77	16.5%

ECB reference rates are quote units per EUR. The model calculates an arithmetic mean over available published business days in each 2025 calendar month and uses that original sale month for both sales and refunds. It does not invert the quote or use today's rate. This is an analytical translation assumption, not the actual transaction rate [S4].

Month	PLN / EUR	CZK / EUR	PLN business days	CZK business days
2025-01	4.2467	25.1633	22	22
2025-02	4.1722	25.0767	20	20
2025-03	4.1820	25.0012	21	21
2025-04	4.2652	25.0389	20	20
2025-05	4.2538	24.9229	21	21
2025-06	4.2658	24.8043	21	21
2025-07	4.2541	24.6246	23	23
2025-08	4.2613	24.5166	21	21
2025-09	4.2589	24.3468	22	22
2025-10	4.2488	24.3154	23	23
2025-11	4.2376	24.2340	20	20
2025-12	4.2239	24.2593	21	21

4. Hub options and scenario economics

The client policy models annual incremental contribution after recurring fixed cost as $C \times u + U \times (1+u) \times s - F$. Capex is year-zero and separate. Payback is simple undiscounted $K / \text{positive annual increment}$; negative or zero increments have no payback. Feasible pairs add country figures without synergy and must fit €450,000 capex and seven FTE [S7].

Option	Capex	FTE	Low	Base	High	Stress	Base payback	Stress payback
NLD+ESP	€270,000	6	€134,532	€269,815	€405,099	€210,514	1.00 yrs	1.28 yrs
CZE+ESP	€225,000	5	€142,119	€286,167	€430,215	€120,247	0.79 yrs	1.87 yrs

Feasible pair ranking (base case)

Feasible pair	Base increment	Stress increment	Capex	FTE	Base payback
CZE+ESP	€286,167	€120,247	€225,000	5	0.79 yrs
POL+ESP	€277,483	€120,418	€240,000	6	0.86 yrs
NLD+ESP	€269,815	€210,514	€270,000	6	1.00 yrs
POL+CZE	€254,747	€-2,818	€205,000	5	0.80 yrs
NLD+CZE	€247,542	€87,741	€235,000	5	0.95 yrs
NLD+POL	€238,208	€87,261	€250,000	6	1.05 yrs
FRA+ESP	€224,899	€168,154	€340,000	7	1.51 yrs
FRA+CZE	€203,483	€46,238	€305,000	6	1.50 yrs
FRA+POL	€194,101	€45,711	€320,000	7	1.65 yrs
FRA+NLD	€184,456	€133,829	€350,000	7	1.90 yrs
DEU+CZE	€182,005	€25,124	€375,000	7	2.06 yrs

The four DEU pairs with FRA, NLD, POL and ESP are not feasible under the stated caps; they are retained as explicit rejections in the workbook and scenario metadata. The selected NLD+ESP pair is not the base-case maximum, but it is the highest defined-stress feasible pair among alternatives positive in all four cases. Its stress case avoids the policy's additional 10% net-sales shock for PLN/CZK markets.

Interpretation guardrail

These are scenario arithmetic outputs, not a statistical confidence interval, causal elasticity estimate or discounted cash-flow model. The decisive unmeasured assumption is local response: the modeled volume uplift and savings must be validated against actual service-lane evidence before full capex release.

5. Recommendation and 90-day implementation

Recommendation: approve **NLD+ESP** with a staged release. Commit to the option envelope (€270,000 capex, six FTE) only as a controlled test. Base economics are €269,815 annual incremental contribution and 1.00-year simple payback; the defined stress remains €210,514 with 1.28-year payback. The strongest base alternative CZE+ESP is €16,352 higher in base annual increment but €90,267 lower in stress increment, reflecting the client policy's CZK shock.

Window	Owner	Actions and dependencies	Decision gate / output
Days 0–30 Frame	COO / Network lead Finance / Data	Validate country-lane SLA baseline, order density, return reasons and actual local carrier/lease/labor quotes. Build a lane-level hub baseline; dependency: carrier, site and labor data.	Gate 1: signed data dictionary, site shortlist, quote pack and KPI baseline. Stop if modeled uplift cannot be tied to an observable lane problem.
Days 31–60 Design	Ops / HR / IT Procurement / Legal	Design inventory pool, WMS/RMA flows, staffing, training, tax/compliance and carrier handoffs. Dependency: site economics, systems integration and local compliance review.	Gate 2: approve or reject NLD+ESP envelope; no lease/capex release without site quote, service case and risk owner.

Window	Owner	Actions and dependencies	Decision gate / output
Days 61–90 Pilot	Hub GM / Quality Commercial	Soft-launch one validated lane/hub first; stage second site only after data shows service and contribution signal. Run weekly control tower with Finance and Data.	Gate 3: release second-hub spend only if 8-week run-rate achieves $\geq 80\%$ of base increment for the pilot scope, OTIF $\geq 95\%$, stockout $\leq 3\%$, and return rate $\leq$ baseline +1pp.

KPI plan

KPI	Definition	Initial target / use
OTIF	On-time, in-full dispatch for hub-served orders	$\geq 95\%$ through pilot; gate metric
Order-to-ship	Median hours from order release to dispatch	$\leq 24\text{h}$ or $\geq 20\%$ improvement vs lane baseline
Contribution	Net sales – net COGS – fulfillment, after actual recurring cost	$\geq 80\%$ of modeled base run-rate before second-hub release
Return rate	Returned units / shipped units by hub-served lane	No more than baseline +1 percentage point
Stockout / fill	Orders delayed by inventory availability	Stockout $\leq 3\%$; fill $\geq 95\%$
Capex / staffing	Committed capex and filled FTE vs envelope	$\leq €270\text{k}$ and ≤ 6 FTE unless board re-approves

Key risks and mitigations

- Demand/uplift risk — validate order density and SLA pain by lane; use the 90-day gate and defer if evidence is weak.
- FX / return risk — keep PLN/CZK exposure explicit; monitor refund rate using the original sale-month rate and run monthly stress refreshes.
- Fixed-cost and labor risk — require signed quotes and staged hiring; do not convert simple payback into a promise.
- Data / process risk — preserve order-level revision and RMA lineage; reconcile weekly hub KPIs to the source-room definitions.

6. Methods, limitations and source register

Calculation method

For each eligible order, gross local sales = quantity × unit price – discount; local gross sales and explicit summed refunds are divided by the original sale month’s mean ECB local-per-EUR quote. Unit cost uses the latest effective-dated EUR/unit record on or before shipment. Net COGS = gross COGS – restocked quantity × original unit cost. Gross sales, refunds, gross/recovered/net COGS, fulfillment and contribution are rounded per order to cents using half-up rounding, then summed. Country and monthly margins are null only where net sales is zero.

Limitations

- Synthetic order/return history and hub assumptions are not measured demand or causal evidence.
- ECB reference rates are analytical translation assumptions, not the actual transaction rates or hedging outcomes.
- Scenario arithmetic is undiscounted and excludes taxes, working capital, ramp timing, shared fixed costs, and synergy; pairs are additive with no synergy.
- World Bank GDP per capita is current US\$, not PPP or constant-price real income, and neither GDP nor population proves product demand.

Source register and links

The full machine-readable register is in evidence/source-register.json and the workbook Sources tab. Key links:

ID	Source	Original URL / file	Use
S1	Client data dictionary	evidence/raw/data-dictionary.md	Synthetic client metadata; grain, revisions, FX and accounting rules.
S2	Client orders, corrections and returns	evidence/raw/orders-part1.csv; evidence/raw/orders-part2.csv; evidence/raw/order-corrections.csv; evidence/raw/returns.csv	Synthetic client exports; eligible 2025 orders and cutoff returns.
S3	Client unit costs and hub options	evidence/raw/unit-costs.csv; evidence/raw/hub-options.csv	Synthetic assumptions; effective-dated unit cost and option economics.
S4	ECB historical reference rates	https://www.ecb.europa.eu/stats/eurofxref/eurofxref-hist.zip	Official archived snapshot; quote units per EUR; source-room retrieval 2026-09-27.
S5	World Bank population archive	https://api.worldbank.org/v2/country/DEU;FRA;NLD;POL;CZE;ESP/indicator/SP.POP.TOTL?date=2022:2024&format=json&per_page=1000	Official archived snapshot; population persons; lastupdated 2026-07-13.
S6	World Bank GDP-per-capita archive	https://api.worldbank.org/v2/country/DEU;FRA;NLD;POL;CZE;ESP/indicator/NY.GDP.PCAP.CD?date=2022:2024&format=json&per_page=1000	Official archived snapshot; current US\$ per person; lastupdated 2026-07-13.
S7	Scenario policy	evidence/raw/scenario-policy.md	Synthetic board assumptions: caps, uplift cases and stress formula.

Reproducibility

Run python scripts/analyze.py to rebuild analysis/processed, evidence/source-register.* and deliverables/metrics.json from the preserved raw files. Run python scripts/create_deliverables.py to rebuild this report, the board presentation and workbook. Run python scripts/verify_outputs.py for schema, reconciliation, scenario, workbook and PDF checks. The raw files, scripts and instructions stay in this project workspace.

End of report